

Homework 15: Rotations in Higher Dimensions

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BCS A, E, F, G

Introduction.

While projections and reflections work regardless of the dimension of the space that they act on, extending rotations to dimensions higher than 2 is somewhat delicate. For example, in $\mathbb{R}^3$ it does not make sense to say something like “rotate $\mathbf{v} = (1,2,3)$ counter-clockwise by an angle of $\pi/4$ ”—this instruction is under-specified, since *one angle alone does not tell us in which direction we should rotate*.

In $\mathbb{R}^3$, we can get around this problem by specifying an axis of rotation via a unit vector (much like we used a unit vector to specify which line to project onto or reflect through earlier). However, *we still have the problem that a rotation that looks clockwise from one side looks counter-clockwise from the other*. Furthermore, there is a slightly simpler method that extends more straightforwardly to even higher dimensions—repeatedly rotate in a plane containing two of the coordinate axes.

Around the x –axis. To illustrate how to do this, consider the linear transformation R_{yz}^θ that rotates $\mathbb{R}^3$ by an angle of θ around the x –axis, from the positive y –axis toward the positive z –axis. It is straightforward to see that rotating $\mathbf{e}_1$ in this way has no effect at all (i.e., $R_{yz}^\theta(\mathbf{e}_1) = \mathbf{e}_1$). Similarly, to rotate $\mathbf{e}_2$ or $\mathbf{e}_3$ we can just treat the yz -plane as if it were $\mathbb{R}^2$ and repeat the derivation to see that

$$R_{yz}^\theta(\mathbf{e}_2) = \begin{bmatrix} 0 \\ \cos \theta \\ \sin \theta \end{bmatrix}, \quad R_{yz}^\theta(\mathbf{e}_3) = \begin{bmatrix} 0 \\ -\sin \theta \\ \cos \theta \end{bmatrix}.$$

It follows that the standard matrix of R_{yz}^θ is

$$R_{yz}^\theta = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix}.$$

A similar derivation shows that we can analogously rotate around the other two coordinate axes in $\mathbb{R}^3$ via the standard matrices

$$R_{zx}^\theta = \begin{bmatrix} \cos \theta & 0 & -\sin \theta \\ 0 & 1 & 0 \\ \sin \theta & 0 & \cos \theta \end{bmatrix}, \quad R_{xy}^\theta = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Remark. We list R_{zx}^θ here instead of R_{xz}^θ to keep $-\sin \theta$ in the upper triangular part. R_{xz}^θ is the same but with the signs of the off-diagonal terms swapped.

Problem 1. Rotate $\mathbf{v} = (3, -1, 2)$ around the z –axis by an angle of $\theta = 2\pi/3$ in the direction from the positive x –axis to the positive y –axis.

Definition. Composition of Linear Transformations.

Suppose $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $S: \mathbb{R}^m \rightarrow \mathbb{R}^p$ are linear transformations. Then their composition is the function

$$S \circ T: \mathbb{R}^n \rightarrow \mathbb{R}^p$$

defined by

$$(S \circ T)(\mathbf{v}) = S(T(\mathbf{v})), \quad \text{for all } \mathbf{v} \in \mathbb{R}^n.$$

Theorem. Composition of Linear Transformations.

Suppose $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $S: \mathbb{R}^m \rightarrow \mathbb{R}^p$ are linear transformations with standard matrices $T \in M_{m,n}$ and $S \in M_{p,m}$ respectively. Then

$$S \circ T: \mathbb{R}^n \rightarrow \mathbb{R}^p$$

is a linear transformation, and its standard matrix is

$$S \circ T = ST.$$

Problem 2. Find the standard matrix of the linear transformation T that reflects $\mathbb{R}^2$ through the line $y = 4/3x$ and then stretches it in the x –direction by a factor of 2 and in the y –direction by a factor of 3 and then rotates it through an angle of π radians.

Problem 3. Construct a Rotation Around a Particular Line in $\mathbb{R}^3$.

Let $R_{\mathbf{u}}^{\theta}$ be the linear transformation that rotates $\mathbb{R}^3$ around the unit vector $\mathbf{u}$ clockwise (when looking in the direction that $\mathbf{u}$ points) by an angle of θ . Its standard matrix is

$$R_{\mathbf{u}}^{\theta} = (\cos \theta)I + (1 - \cos \theta)\mathbf{u}\mathbf{u}^T + \sin \theta \begin{bmatrix} 0 & -u_3 & u_2 \\ u_3 & 0 & -u_1 \\ -u_2 & u_1 & 0 \end{bmatrix}.$$

- a. Compute $R_{\mathbf{u}}^{\theta}$ if $\mathbf{u} = (1,0,0)$. Which matrix that we already saw in the homework introduction's does it equal?
- b. Compute $R_{\mathbf{u}}^{\pi/4}(\mathbf{v})$ if $\mathbf{u} = (1,2,2)/3$ and $\mathbf{v} = (3,2,1)$.
- c. Show that $R_{yz}^{\pi/3} \circ R_{xy}^{\pi/3}$ is a rotation matrix.

[Side note: the composition of any two rotations is always a rotation (even in dimensions higher than 3), but proving this fact is outside of the scope of this course.]

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Good Luck